Gravity as Information Recovery: The Gravitational Refractive Index is the Petz Fidelity Bound

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Three quantities from three fields turn out to be the same number. From general relativity: the coordinate speed of light $c_{\rm eff}/c = e^{\Phi/c^2}$. From thermodynamics: the entropy production cost $e^{-\Sigma_{\rm grav}/2}$ with $\Sigma_{\rm grav} = 2|\Phi|/c^2$. From quantum information: the Petz recovery fidelity bound $F_{\rm bound} \geq e^{-\Sigma/2}$. The identification $c_{\rm eff}/c = e^{-\Sigma_{\rm grav}/2} = F_{\rm bound}$ states that light slowing in a gravitational field *is* information loss, and the gravitational refractive index *is* the inverse recovery fidelity. Three independent routes—modular flow, gravitational Landauer principle, and quantum channel analysis—converge on $\Sigma_{\rm grav} = -\ln(-g_{00})$. The identity $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$ yields the exponential metric $g_{00} = -e^{-r_{\rm s}/r}$, which has no event horizon ($\tau < 1$ everywhere) and agrees with Schwarzschild through second post-Newtonian order. Strong-field deviations—4.6% larger shadow, $\sim 4\%$ shifted quasinormal modes, and gravitational-wave echoes at $\Delta t \approx 4\, r_{\rm s}/c$—are within reach of current observatories.

INTRODUCTION

In 1911, Einstein observed that the speed of light varies in a gravitational field [1]. Dicke later showed that general relativity (GR) can be exactly reformulated as flat-spacetime optics with a gravitational refractive index [2]: in a static field with Newtonian potential $\Phi < 0$,

$$\frac{c_{\rm eff}}{c} = e^{\Phi/c^2}\,. \tag{1}$$

Meanwhile, quantum information theory has established that any quantum channel $\mathcal{N}$ with entropy production $\Sigma \equiv D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ admits a universal recovery bound [3, 4]:

$$F\big(\rho,\ \mathcal{R}_\sigma \circ \mathcal{N}(\rho)\big)\ \geq\ e^{-\Sigma/2}\,, \tag{2}$$

where $\mathcal{R}_\sigma$ is the (rotated) Petz recovery map [3] and F is the Uhlmann fidelity. The Petz map is the *unique* retrodiction functor satisfying Bayesian consistency [5]—the quantum generalization of Bayes' rule. In a companion paper [6], we showed that the temporal asymmetry parameter $\tau = 1 - F \in [0, 1]$ provides an operational measure of the arrow of time, with $\tau = 0$ for closed systems and $\tau \leq 1 - e^{-\Sigma/2}$ in general.

Here we connect these two results. Equations (1) and (2) share the same exponential structure. This is not a coincidence: they are the same physical statement in different languages.

THE TRIPLE IDENTIFICATION

We define the dimensionless gravitational entropy production as

$$\Sigma_{\rm grav}\ \equiv\ \frac{2|\Phi|}{c^2}\ =\ \frac{r_{\rm s}}{r}\,, \tag{3}$$

where $r_{\rm s} = 2GM/c^2$ is the Schwarzschild radius. This is an *informational* entropy production—the cost of retrodiction from infinity to radius r—not a Clausius entropy. The Clausius entropy production for a photon in Tolman thermal equilibrium vanishes, since $T(r)\sqrt{-g_{00}} = T_\infty$ [7]. The quantity $\Sigma_{\rm grav}$ persists even in equilibrium: it measures the informational cost of climbing the gravitational potential.

Three independent arguments converge on Eq. (3), all sharing the mathematical core $\Sigma_{\rm grav} = -\ln(-g_{00})$:

(a) Modular flow [known].—In algebraic QFT on Schwarzschild, the quantum relative entropy (QRE) between the vacuum and a coherent excitation is computable via modular theory [8]. The Tolman blueshift and energy redshift together give a fractional QRE loss $(D_{\rm in} - D_{\rm out})/D_{\rm in} = r_{\rm s}/r$, exactly.

(b) Gravitational Landauer principle [known].—The Tolman-modified cost of erasing one bit at radius r exceeds the cost at infinity by $1/\sqrt{-g_{00}}$ [9]. The dimensionless excess erasure cost yields $\Sigma_{\rm grav} = -\ln(-g_{00})$, which equals $r_{\rm s}/r$ for the exponential metric and $-\ln(1 - r_{\rm s}/r) \approx r_{\rm s}/r + O((r_{\rm s}/r)^2)$ for Schwarzschild.

(c) Quantum channel [known].—Modeling gravitational redshift as a pure-loss bosonic channel with intensity transmissivity $\eta = -g_{00}$, the QRE decrease is $\Sigma = -\ln\eta = r_{\rm s}/r$ [10]. Petz recovery for this channel type has been experimentally demonstrated [30].

No route constructs a canonical gravitational CPTP map with explicit Kraus operators [**?**]. We bypass this: any consistent classical-gravity–quantum-matter coupling is irreversible [28], guaranteeing the existence of an irreversible channel; route (a) fixes the coefficient; route (c) provides a concrete existence proof. The modular channel of Trejo-Calderón [**?**] is the most promising candidate for a first-principles construction.

Substituting Eq. (3) into the Petz bound (2) gives

$F_{\rm bound} = e^{-\Sigma_{\rm grav}/2} = e^{\Phi/c^2}$. Comparing with Eq. (1):

$$\boxed{\frac{c_{\rm eff}}{c} \;=\; e^{-\Sigma_{\rm grav}/2} \;=\; F_{\rm bound}\,.} \tag{4}$$

This is the central result [new synthesis]: the coordinate speed of light *is* the Petz recovery fidelity bound. Equivalently, the gravitational refractive index is the inverse fidelity: $n_{\rm grav} = c/c_{\rm eff} = 1/F_{\rm bound} = e^{+\Sigma_{\rm grav}/2}$.

The physical picture is transparent. An alternative route to this identification uses quantum hypothesis testing [26]: $\Sigma_{\rm grav} = D(\rho(r_1)\|\rho(r_2))$, where $\rho(r)$ is the quantum state of a probe at radius r. This definition bypasses the need for an explicit CPTP map and gives $\Sigma_{\rm grav} = -\ln(-g_{00})$ directly from the Kubo–Martin–Schwinger (KMS) state structure in curved spacetime.

A photon climbing out of a gravitational potential loses information—its quantum state degrades due to interaction with the gravitational channel. The recovery fidelity $F_{\rm bound}$ quantifies how well the original state can be retrodicted. At the would-be event horizon, $F_{\rm bound} \to 0$ and $c_{\rm eff} \to 0$: complete information loss coincides with the vanishing of the coordinate speed of light. The event horizon *is* an information horizon, literally.

Table I verifies the identification across four orders of magnitude in gravitational compactness. The agreement is exact for the exponential metric and holds through $O(r_{\rm s}/r)$ for Schwarzschild; the first discrepancy is at $O((r_{\rm s}/r)^3)$, which is $\sim 10^{-18}$ in the solar system [11]—far below measurability.

CONSEQUENCES

Model-independent no-horizon theorem.—The Petz bound alone, before specifying $\Sigma_{\rm grav}$, has a consequence: if gravitational redshift is a quantum channel with *finite* entropy production, then $F \geq e^{-\Sigma/2} > 0$, so $\tau < 1$. An event horizon requires $F = 0$ (complete information loss), which demands $\Sigma \to \infty$—impossible for a finite-mass object at finite distance. The Petz bound forbids event horizons model-independently [4]; the specific form of $\Sigma_{\rm grav}$ is not needed.

The exponential metric.—The identity $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$, taken as exact, determines the metric uniquely [12, 13]:

$$ds^2 = -e^{-r_{\rm s}/r}\,c^2dt^2 + e^{+r_{\rm s}/r}\big(dr^2 + r^2d\Omega^2\big)\,, \tag{5}$$

in isotropic coordinates. Since $e^{-r_{\rm s}/r} > 0$ for all $r > 0$, there is no event horizon: $\tau < 1$ everywhere. This metric has a long independent pedigree: Papapetrou [12], Yilmaz [13], Rosen's bimetric theory [14], and three scalar-field families within GR that all reduce to it when scalar charge equals gravitational mass [15]. Boonserm et al. classified it as a traversable wormhole with throat at $R_{\rm thr} = (e/2)\,r_{\rm s}$ [16]. It shares PPN parameters $\gamma = \beta = 1$ with Schwarzschild and passes all solar-system tests [11]. Our contribution is the information-theoretic motivation: the exponential metric is the geometry whose redshift factor *equals* the Petz recovery fidelity bound, $\sqrt{-g_{00}} = e^{-\Sigma/2}$, taken as an exact identity rather than a one-sided inequality. The spatial component $g_{rr} = e^{+r_{\rm s}/r}$ follows from the isometry condition $g_{00}\cdot g_{rr} = -1$ (Dicke's scalar refractive index [2]), which is an independent assumption—information theory constrains only g_{00}. Predictions are thus *robust* (no-horizon theorem, $n_{\rm grav} = 1/F_{\rm bound}$, decoherence bound—depending only on g_{00}) or *conditional* (shadow, QNM, echoes—depending on the full metric).

Caveat on saturation.—The JRSWW bound [4] is structurally *strict* for $\Sigma > 0$: the proof proceeds via the Golden–Thompson inequality, which is an equality only when the relevant operators commute—precisely the $\Sigma = 0$ case [3]. No known quantum channel saturates the bound for $\Sigma > 0$. In weak fields, the gap is $O((r_{\rm s}/r)^2)$, far below current measurability. The exponential metric (5) should therefore be understood not as the unique consequence of saturation, but as the geometry defined by the *identity* $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$, which holds independently of whether the Petz bound is tight. Whether nature selects this geometry (exponential metric) or admits a gap (a metric between exponential and Schwarzschild) is empirical. The exponential metric is *not* a vacuum solution of the Einstein equations; it requires a phantom scalar source [15, 16]. We motivate the *metric* from information theory, not field equations—an important distinction from the Yilmaz program [13], whose modified field equations face well-documented difficulties [17].

Gravitational decoherence.—Pikovski et al. [18] showed that gravitational time dilation universally dephases composite quantum systems in spatial superposition. In the τ framework [6], the resulting temporal asymmetry is $\tau_{\rm grav}(t) = 1 - e^{-N\Gamma t^2}$, where $\Gamma \propto (\Delta E)^2(\Delta\Phi)^2/(\hbar^2c^4)$. The dephasing rate is controlled by $\Delta\Phi/c^2 \sim \Sigma_{\rm grav}/2$: the gravitational entropy production governs both the static (refractive index) and dynamic (decoherence) aspects of gravitational information loss. Bonder et al. [19] noted that the Pikovski effect is frame-dependent; our framework is consistent with this, since τ depends on the observer's channel, not on an absolute decoherence rate.

TABLE I. Numerical verification of the triple identification. Φ/c^2 is the dimensionless potential, $\Sigma_{\rm grav} = 2|\Phi|/c^2$, and $F_{\rm bound} = e^{-\Sigma_{\rm grav}/2} = c_{\rm eff}/c$.

Location	Φ/c^2	$\Sigma_{\rm grav}$	$F_{\rm bound}$	$c_{\rm eff}/c$
Earth	-7×10^{-10}	1.4×10^{-9}	$1-7\times10^{-10}$	$1-7\times10^{-10}$
Sun	-2.1×10^{-6}	4.2×10^{-6}	$1-2.1\times10^{-6}$	$1-2.1\times10^{-6}$
NS	-0.2	0.4	0.819	0.819
Horizon	$-\infty$	∞	0	0

RELATION TO PRIOR WORK

The logic here is complementary to the Jacobson program [20]. Jacobson derives *field equations* from thermodynamic entropy applied to local Rindler horizons: $\delta Q = T\delta S \to$ Einstein equations $\to$ Schwarzschild. We derive the *metric component* directly from information recoverability: $F_{\rm bound} = e^{-\Sigma_{\rm grav}/2} \to g_{00} \to$ exponential metric. Both programs share the premise that geometry is information-theoretic; they differ in what is derived first (field equations vs. metric) and consequently arrive at different strong-field predictions. The Dorau–Much result [8]—deriving the semiclassical Einstein equations from quantum relative entropy via modular theory—suggests that the two programs share a common informational foundation and may ultimately converge.

In holographic settings, the Petz map reconstructs bulk quantum fields from boundary data [21], with fidelity $F_{\rm bulk} \geq e^{-\Delta S_{\rm gen}/2}$, where $\Delta S_{\rm gen}$ is the change in generalized entropy. Our identification (4) extends this structure beyond AdS/CFT: the bulk metric *itself* encodes the recovery fidelity, with $\sqrt{-g_{00}}$ playing the role of $F_{\rm bound}$. The Crooks fluctuation theorem has recently been established in curved spacetimes [22], confirming that the Crooks–Petz connection of Paper I [6] extends to general-relativistic settings—entropy production in curved spacetime is observer-dependent and quantified by $\Sigma_{\rm grav}$.

Galley, Giacomini, and Selby [28] proved that *any* consistent coupling between classical gravity and quantum matter is fundamentally irreversible ($\Sigma > 0$), providing model-independent support for the premise that gravitational processes produce entropy. Bai, Buscemi, and Scarani [27] independently constructed a fully quantum entropy production using Bayesian retrodiction—mathematically parallel to the τ framework but without gravitational application. De Vuyst et al. [29] showed via quantum reference frames that gravitational entropy is observer-dependent, providing algebraic justification for the coordinate dependence of $\Sigma_{\rm grav}$.

TESTABLE PREDICTIONS

The exponential and Schwarzschild metrics agree through second post-Newtonian order but diverge in strong fields, yielding three testable predictions ordered by near-term feasibility.

(1) Shadow size.—The exponential photon sphere at $r_{\rm ph} = r_{\rm s}$ (isotropic coordinates) yields a critical impact parameter $b_{\rm crit} = 2e\,m \approx 5.44\,m$ versus Schwarzschild's $3\sqrt{3}\,m \approx 5.20\,m$ ($m = GM/c^2$): a 4.6% larger shadow. For M87*, this gives 41.5 μas versus 39.7 μas; the EHT measurement of 42±3 μas [23] is consistent with both but closer to the exponential value. Next-generation EHT, targeting sub-μas precision, can distinguish the two.

(2) Quasinormal modes.—In the eikonal limit, $f^{\rm exp}_{\rm QNM}/f^{\rm Schw}_{\rm QNM} = 3\sqrt{3}/(2e) \approx 0.956$, a $\sim 4.4\%$ downward shift. This estimate should be treated with caution: the exponential photon sphere coincides with the wormhole throat, where the effective potential differs qualitatively from Schwarzschild. WKB calculations for a generalized exponential wormhole find a -7.8% shift [24]. Current LIGO ringdown analyses constrain QNM deviations at the $\sim$ 5–10% level; next-generation detectors (Einstein Telescope, Cosmic Explorer) will reach $\sim 1\%$ precision.

(3) Gravitational-wave echoes.—The most distinctive signature. The exponential metric's traversable wormhole structure produces echoes—repeated, damped pulses following the initial ringdown—with time delay

$$\Delta t_{\rm echo} \approx \frac{4.2\, r_{\rm s}}{c}\,, \tag{6}$$

giving $\sim$ 2.5 ms for a $60\,M_\odot$ remnant at $\sim$ 400 Hz, squarely in the LIGO band. This is $\sim 90\times$ shorter than Planck-wall models [25], because the exponential metric has no horizon and its tortoise coordinate $r^* = \int e^{r_{\rm s}/r}\,dr$ remains bounded. Current echo searches use $\sim$ 200 ms templates; dedicated short-delay searches are needed to test this prediction.

SCOPE AND LIMITATIONS

The identification (4) *can*: unify quantum error correction, thermodynamic irreversibility, and gravitational physics under a single parameter τ; provide universal recovery bounds that hold independently of the specific protocol; connect the Hayden–Preskill decoder—which *is* the Petz map [21]—to gravitational physics; and provide a precise, computable definition of the gravitational arrow of time: $\tau > 0$ whenever $\Sigma_{\rm grav} > 0$, with $\tau = 0$ if and only if spacetime is flat.

It *cannot*: explain dark matter ($\Sigma_{\rm grav} \sim 10^{-6}$ at galactic scales, far too small); Bianconi's gravity-from-entropy framework [32] yields Schwarzschild with $O(r^{-4})$ corrections in vacuum [31], *not* the exponential metric; our information-theoretic derivation via $\Sigma_{\rm grav} = -\ln(-g_{00})$ is independent of that program. It cannot explain the cosmological constant or dark energy; derive the full metric tensor g_{ab} from τ (the relation $g_{00} \to \Sigma_{\rm grav} \to \tau$ is one-directional; τ constrains only g_{00}, not the spatial components); or extend to general dynamical spacetimes without modification. A promising extension uses the Kodama vector [**?**]—the unique divergence-free vector in spherically symmetric spacetimes—giving $\tau_K(R) = 1 - \sqrt{1 - H^2R^2}$ in FRW, tracking apparent (not event) horizons; the deeper foundation is modular flow via crossed-product algebras [**? ?**].

CONCLUSION

The coordinate speed of light, the informational entropy production cost, and the Petz recovery fidelity bound are the same number: $c_{\rm eff}/c = e^{-\Sigma_{\rm grav}/2} = F_{\rm bound}$. This is a new connection, not a new theory—each piece is known; the synthesis is new. The exponential metric follows from the identity $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$, with testable strong-field deviations from Schwarzschild. The no-horizon theorem holds model-independently: for any finite $\Sigma_{\rm grav}$, $\tau < 1$ and complete information loss cannot occur—regardless of whether the Petz bound is tight or strict.

The deepest implication is that the metric tensor does not merely *resemble* an information-theoretic quantity—$\sqrt{-g_{00}}$ *is* the recovery fidelity bound. Whether nature saturates this bound is a question that the strong-field observational program of the coming decade can answer.

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